

A054-v0.4.0-r3 Mechanism Injection Falsifier

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1 Identification

A054-v0.4.0-r3; thin instance of SST Falsifier Framework v1.0.4 CANONICAL_FROZEN.

2 Question

Can a preregistered reduced closure turn frozen coherent-but-nonconfining A054 geometries into restoring, Kelvin-bounded and nonlinear-ringdown-bounded branches without per-candidate retuning?

3 Hypotheses

H_0 : no registered mechanism transfers to held-out geometries under unchanged v0.2 hard gates. H_1 : at least one preregistered arm and global gain does.

4 Assumptions

CORE and ELASTIC are reduced effective closures and are not first-principles Euler pressure or shear-modulus derivations. Geometry semantics are irrelevant to blind scoring.

5 Symbols

a is core radius, g global dimensionless gain, $\mathbf{v}_{BS}$ baseline velocity, $\mathbf{w}_P$ and $\mathbf{w}_E$ reduced mechanism fields.

6 Equations

Machine-readable science contract

Research question Can any preregistered reduced mechanism convert the frozen A054 coherent-but-nonconfining three-filament states into restoring, Kelvin-bounded and nonlinear-ringdown-bounded branches without per-candidate retuning or relaxing the v0.2 hard gates?

Objective Identify the minimal registered closure class, if any, that produces held-out dynamical recovery under the original A054-v0.2 thresholds while preserving topology and numerical convergence.

H_0 H_0 : none of the registered mechanism arms recovers a held-out and numerically certified dynamically confined branch beyond the Biot-Savart baseline.

H_1 H_1 : at least one preregistered mechanism arm and one globally selected gain recovers held-out dynamically confined branches under unchanged v0.2 gates.

Registered equations

E1: Baseline regularized Biot–Savart velocity

$$\mathbf{v}_{BS}(\mathbf{x}) = \sum_j \frac{\Gamma_j}{4\pi} \oint_{C_j} \frac{d\mathbf{l} \times (\mathbf{x} - \mathbf{y})}{|\mathbf{x} - \mathbf{y}|^3} \quad (1)$$

Dimensional check: $\Gamma_j [L^2/T]$ times $d\mathbf{l} [L]$ times $\mathbf{r} [L]$ over $length^3$ gives L/T .

E2: Finite-core cohesive-stress surrogate

$$\mathbf{F}_P(\mathbf{x}_i) = \sum_{j \in \text{nei}} \left[\frac{r_{ib}^2}{(2\sigma_r^2)} \sigma_r^2 - \chi \frac{r_{ib}^2}{(2\sigma_a^2)} \sigma_a^2 \right] (\mathbf{x}_i - \mathbf{x}_j) \quad (2)$$

Dimensional check: The raw field scales as L^{-1} ; it is converted to L/T only by registered RMS normalization E4.

E3: Curvature-elastic surrogate

$$E_B = \frac{1}{2} \sum_i \int_{C_i} \kappa^2 ds, \quad \mathbf{f}_E = \mathbf{f} \times (-\nabla_{\perp} E_B) \quad (3)$$

Dimensional check: The discrete variational field is a raw geometric field and is converted to L/T only by E4.

E4: Candidate-independent mechanism normalization and injection

$$\widehat{\mathbf{w}}_m = V_0 \mathbf{w}_m / \text{RMS}(\mathbf{w}_m|_{t=0}), \quad \dot{\mathbf{X}} = \mathbf{v}_{BS} + \mathbf{g}_s \quad (4)$$

Dimensional check: V_0 has L/T ; normalized mechanism fields and total $\dot{\mathbf{X}}$ therefore have L/T .

E5: Hard recovery criterion inherited unchanged from A054-v0.2

$$R_{\text{restore}} \leq 1.25, \quad G_K \leq 0.12, \quad A_{\text{ring}}/A_0 \leq 1.75, \quad \Delta L_K \leq 0.05 \quad (5)$$

Dimensional check: All registered recovery statistics are dimensionless.

Registered code-to-mathematics steps

1. **S1** [G0; equations E1]: Verify frozen protocol, public-source blindness and exact input hashes before numerical work.
2. **S2** [G1; equations E1]: Load exact frozen v0.2 anonymous discovery geometries and opaque source-native confirmation geometries.
3. **S3** [G2; equations E1, E2, E3, E4]: Validate analytic null behavior, rigid-objectivity and mechanism-field normalization on the Python FP64 reference.
4. **S4** [G3; equations E4, E5]: Blindly evaluate BASE, CORE, ELASTIC and CORE+ELASTIC arms over the preregistered gain grid on discovery; select one global gain per arm without semantic labels.
5. **S5** [G4; equations E4, E5]: Apply each selected arm/gain unchanged to the held-out opaque source-native confirmation cohort.

6. **S6** [G5; equations E5]: For confirmed recovery cells, require resolution convergence and C++/OpenMP FP64 parity; RPO/Floquet is evaluated only after restoring/Kelvin/ringdown recovery.
7. **S7** [G6; equations E4]: Optional GPU screening remains non-authoritative.
8. **S8** [G7; equations E5]: Assess evidence-class eligibility and mechanism minimality without promoting simulation-only evidence to physical support.

7 Algorithm

Frozen-source verification; reference checks; blind discovery; no-refit held-out confirmation; full numerical/native certification; conditional RPO/Floquet only after hard recovery.

8 Sources

Discovery is the exact 72-case v0.2 FULL blind geometry manifest. Confirmation is seven source-native three-component KnotPlot links encoded under opaque IDs.

9 Gates

G0–G9 use the framework dependency ledger; physical promotion is prohibited without eligible G7 replication.

10 Numerics

Python/NumPy FP64 is reference arithmetic. Original v0.2 hard thresholds remain frozen. Active mechanism gains are global, never case-specific.

11 Backends

C++17/OpenMP FP64 certifies instance mechanism kernels. SYCL FP32/DD32 is optional screening-only under framework policy.

12 Blindness

Confirmation topology names and particle semantics are private and nonce-commitment bound. Public scoring uses only opaque IDs.

13 Statistics

Arm/gain selection is lexicographic on recovery count and a preregistered secondary continuous score; confirmation is a strict held-out transfer with no refit.

14 Results

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	b4f667c3af0a262459da65768e65bfa2acc4dbe04374751ebfc128dc68b464b9
Framework schema	SST-FALSIFIER-PROTOCOL-2

Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	7c5c29dbeb1314dd47f95d066483e9e3d87e7b14d06c793418863acb2edcdfd9

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen v0.4 protocol intact and is runtime provenance self-consistent?
G1	PASS	Are the exact discovery and held-out confirmation sources present, has the
G2	PASS	Do reference numerics, dimensions, objectivity and mechanism-null checks
G3	FAIL	Does blind discovery identify at least one registered mechanism arm with
G4	NOT_RUN_PREREQUISITE	Does the globally selected arm/gain transfer without refit to held-out s
G5	NOT_RUN_PREREQUISITE	Do confirmed recoveries survive convergence and strict C++/OpenMP
G6	NOT_RUN_PREREQUISITE	If explicitly enabled, does optional GPU screening satisfy CPU-to-GPU
G7	NOT_RUN_PREREQUISITE	Does certified support replicate across eligible physical evidence classes
G8	NOT_RUN_PREREQUISITE	Does the registered mechanism/ablation logic uniquely identify the min
G9	DEFERRED	After blind gates, do reveal and forbidden-term commitments verify ex

Backend manifest

```
{
  "status": "NO_REGISTERED_MECHANISM_RECOVERS_DISCOVERY",
  "winner": {
    "arm": "ELASTIC",
    "cell_count": 48,
    "gain": 0.25,
    "median_log_restore_plus_ringdown": 5.414237900977529,
    "recovered_cells": 0
  }
}
```

Source manifest

```
{
  "manifest_sha256": "64eed9962d8c109aab9564899420db5cb64b500de0007a6d9d35d6d56eff9c4",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A054",
      "evidence_class": "simulation",
      "exists": false,
      "independence_family": "A054_V020_FROZEN_BLIND_GEOMETRY",
      "path": "C:\\workspace\\projects\\SST-Workbench\\data\\discovery\\DISCOVERY_MANIFEST.json",
      "provenance_family": null,
      "registry_hint": {
        "catalog_id": "A054",
        "hits": []
      },
      "role": "discovery",
      "source_id": "D0",
      "version": "v0.2.0"
    }
  ],
}
```

```

{
  "catalog_id": "A054",
  "evidence_class": "simulation",
  "exists": false,
  "independence_family": "KNOTPLOT_SOURCE_NATIVE_LINK_EXPORTS",
  "path": "C:\\workspace\\projects\\SST-Workbench\\data\\confirmation\\CONFIRMATION_MANIFEST",
  "provenance_family": null,
  "registry_hint": {
    "catalog_id": "A054",
    "hits": []
  },
  "role": "confirmation",
  "source_id": "C0",
  "version": "v0.3.0-source-native"
}
],
"workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}

```

15 Interpretation

A simulation-level recovery identifies a reduced mechanism candidate. It does not establish nucleon identity or a first-principles SST closure.

16 Reproducibility

Framework pin, frozen protocol, source SHA-256 manifests, blind commitments, environment/backend manifests and output manifests define the audit trail.

References

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- [3] R. Klein, A. J. Majda, and K. Damodaran, “Simplified equations for the interaction of nearly parallel vortex filaments,” *Journal of Fluid Mechanics* **288**, 201–248 (1995). <https://doi.org/10.1017/S0022112095001151>.
- [4] O. Iskandarani, “A054 Restoring-Force $\times$ Kelvin/Floquet Nucleon Certification v0.2.0,” SST-Workbench technical falsifier (2026). Local SST-Workbench A054 family.
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